

Amaliy ish № 3

Mavzu: Operatorlarni qayta yuklashga misollar

Ishning maqsadi:

Ushbu ishning maqsadi, C# dasturlash tilida operatorlarni qayta yuklash (operator overloading) texnikasini o'rganish va uni amaliyotda qo'llashni o'rganishdir. Bu orqali talabalarning ob'ektga yo'naltirilgan dasturlash (OOP) tushunchalarini chuqurlashtirish va sinflar uchun maxsus operatorlar yaratish ko'nikmalarini oshirish maqsad qilingan.

Kerakli asbob-uskunalar va jihozlar:

- Kompyuter yoki noutbuk
- Visual Studio yoki boshqa C# muhitini qo'llab-quvvatlaydigan IDE (Integrated Development Environment)
- C# tilining o'rnatilgan versiyasi

Qisqacha nazariy ma'lumotlar:

Operatorlarni Qayta Yuklash (Operator Overloading)

C# tilida operatorlarni qayta yuklash (operator overloading) sinf yoki strukturaga operatorlarning o'ziga xos xatti-harakatlarini aniqlash imkonini beradi. Bu operatorlar yordamida obyektlar o'rtasida matematik va mantiqiy amallar bajarilishi mumkin.

Operatorlarni Qayta Yuklash Qoidolari:

- Operatorlar statik metodlar sifatida e'lon qilinadi.
- Operatorlar e'lon qilingan sinf yoki strukturaning obyektlarini qabul qilishi va qaytarishi kerak.
- Quyidagi operatorlar qayta yuklanishi mumkin: +, -, *, /, %, &, |, ^, <<, >>, ==, !=, <, >, <=, >=, ++, --, true, false.

Masalaning qo'yilishi:

Quyidagi masalada ikkita uch o'lchovli nuqta (3D point) sinfini yaratish va ular uchun +, -, == va != operatorlarini qayta yuklash kerak.

Masalaning dasturi (C# tilida):

1. Sinfni yaratish va operatorlarni qayta yuklash

```
csharp
Копировать код
using System;

namespace OperatorOverloadingApp
{
    // Point3D sinfi
    public class Point3D
    {
        // Fieldlar
        public double X { get; set; }
        public double Y { get; set; }
        public double Z { get; set; }

        // Konstruktor
        public Point3D(double x, double y, double z)
        {
            X = x;
            Y = y;
            Z = z;
        }

        // + operatorini qayta yuklash
```

```

public static Point3D operator +(Point3D p1, Point3D p2)
{
    return new Point3D(p1.X + p2.X, p1.Y + p2.Y, p1.Z + p2.Z);
}

// - operatorini qayta yuklash
public static Point3D operator -(Point3D p1, Point3D p2)
{
    return new Point3D(p1.X - p2.X, p1.Y - p2.Y, p1.Z - p2.Z);
}

// == operatorini qayta yuklash
public static bool operator ==(Point3D p1, Point3D p2)
{
    return p1.X == p2.X && p1.Y == p2.Y && p1.Z == p2.Z;
}

// != operatorini qayta yuklash
public static bool operator !=(Point3D p1, Point3D p2)
{
    return !(p1 == p2);
}

// ToString metodini qayta yuklash
public override string ToString()
{
    return $"({X}, {Y}, {Z})";
}

// Equals va GetHashCode metodlarini qayta yuklash
public override bool Equals(object obj)
{
    if (obj == null || GetType() != obj.GetType())
        return false;

    Point3D p = (Point3D)obj;
    return X == p.X && Y == p.Y && Z == p.Z;
}

public override int GetHashCode()
{
    return (X, Y, Z).GetHashCode();
}
}

```

2. Obyektlar bilan ishlash

csharp

Копировать код

using System;

namespace OperatorOverloadingApp

{

```

class Program
{
    static void Main(string[] args)
    {
        // Point3D obyektlarini yaratish
        Point3D point1 = new Point3D(1.0, 2.0, 3.0);
        Point3D point2 = new Point3D(4.0, 5.0, 6.0);

        // + operatorini ishlatish
        Point3D sum = point1 + point2;
        Console.WriteLine($"Sum: {sum}");

        // - operatorini ishlatish
        Point3D difference = point1 - point2;
        Console.WriteLine($"Difference: {difference}");

        // == va != operatorlarini ishlatish
        bool areEqual = point1 == point2;
        bool areNotEqual = point1 != point2;
        Console.WriteLine($"Are Equal: {areEqual}");
        Console.WriteLine($"Are Not Equal: {areNotEqual}");
    }
}

```

Topshiriqlar:

1. **Qo'shimcha operatorlar:** Point3D sinfi uchun * (ko'paytirish) va / (bo'lish) operatorlarini qayta yuklang.
2. **Murakkab operatorlar:** Kompleks sonlar (ComplexNumber) sinfini yarating va ular uchun +, -, *, / operatorlarini qayta yuklang.
3. **Logik operatorlar:** Boolean (true/false) qiymatlar bilan ishlovchi sinf yarating va u uchun & (va), | (yoki) operatorlarini qayta yuklang.
4. **Inkrement va dekrement operatorlari:** Point3D sinfi uchun ++ va -- operatorlarini qayta yuklang.
5. **Solishtirish operatorlari:** Point3D sinfi uchun <, >, <=, >= operatorlarini qayta yuklang va ularni ishlatish uchun misollar yozing.